Perspectives and Evolution of the Electricity Mix in Spain (2030-2040): Official Plans, Structural Challenges, and Future Scenarios

Introduction to the New National and European Energy Paradigm

The Spanish electricity system is at the epicenter of an unprecedented structural transformation, driven by a combination of climatic, geopolitical, and technological factors. The transition from a historical, centralized, synchronous model highly dependent on imported fossil fuels to a decentralized, asynchronous paradigm dominated by variable renewable energy sources poses technical and financial challenges of exceptional magnitude[1]. The official planning of the Government of Spain aligns with and even exceeds the guidelines established by the European Green Deal, the *Fit for 55* legislative package, and the *REPowerEU* strategic plan, setting a roadmap that demands a complete reengineering of the national infrastructure[1].

The timeframe between 2030 and 2040 is absolutely critical for this transition. This decade represents the window for consolidating massive investments in grid infrastructure, the electrification of sectors traditionally dependent on hydrocarbons, such as mobility and industrial heating, and the development of a mature storage ecosystem that provides firmness to the intrinsic variability of wind and solar resources[6]. At the European level, forecasts indicate that renewable electricity generation will nearly double to supply about 60% of the EU's electricity by 2030, forcing member states to pace their transmission grids to avoid systemic bottlenecks[9].

This report provides a comprehensive analysis of the existing official plans in Spain regarding energy policy, with a meticulous focus on the projection of the future electricity mix for the milestones of 2030, 2035, and 2040. Through the scrutiny of current regulations, current systemic constraints are evaluated, such as the cannibalization of wholesale market prices, the increase in clean energy curtailment, the loss of rotational inertia in the grid, and the economic implications of the nuclear fleet's closure schedule. Finally, based on geopolitical variables, the evolution of the levelized cost of energy, and demand disruptions, three detailed prospective scenarios are proposed to model the possible evolution of the national energy mix over the next two decades.

Regulatory Framework and Official Planning of the Spanish Electricity System

The architecture of Spanish energy policy is based on three major documentary pillars that provide regulatory certainty, establish mandatory goals, and guide capital allocation by market agents. These instruments are the National Integrated Energy and Climate Plan (PNIEC), the

Long-Term Decarbonization Strategy, and the Electricity Transmission Grid Planning[10].

The National Integrated Energy and Climate Plan (PNIEC) 2023-2030

The National Integrated Energy and Climate Plan (PNIEC) operates as the central strategic planning instrument that translates Spain's binding commitments to the European Union into operational and quantifiable domestic targets[1]. The update to this plan, approved by Royal Decree 986/2024 of September 24, 2024, substantially raises the climate and energy ambition compared to its initial version drafted in 2021[1]. This revision responds both to the accelerating maturity of certain clean technologies and to the urgent need to guarantee energy sovereignty following the severe disruptions suffered in international natural gas markets[1].

The macroeconomic objective structured in the PNIEC 2023-2030 aims to mobilize a total investment of close to 308 billion euros by the end of this decade. Of this capital injection, 37% is expected to be allocated directly to the installation of new renewable generation capacities and 28% to comprehensive energy efficiency measures. Government models project that this mobilization will generate a 3.2% increase in Gross Domestic Product and consolidate the creation of over half a million jobs associated with the ecological transition[1].

In the climate dimension, the updated plan pursues an ambitious 32% reduction in Greenhouse Gas emissions compared to 1990 levels, exceeding the 23% stipulated in the 2021 version[1]. It also projects a drastic decrease in external energy dependence, reducing it from 73% recorded in 2019 to 50% in 2030, which would materialize in estimated cumulative savings of 86.75 billion euros derived from not importing fossil fuels[1].

From the perspective of the electricity system, the document designs a highly decarbonized energy matrix. Driven by the electrification of transport, the adoption of heat pumps, and the deployment of new industrial vectors, national electricity demand is estimated to grow by 34% compared to the 2019 baseline, reaching a total volume of 358 TWh[16]. The overall electrification rate of the economy will consequently scale to 35%[19]. To meet this increased demand, the electricity mix stipulates that 81% of generation must come exclusively from renewable sources, while the share of renewables in final gross energy consumption must reach 48%[1].

To achieve this level of coverage, the PNIEC 2023-2030 articulates a massive expansion of the national generation fleet. The goal is to reach a total installed capacity in the electricity sector of 214 GW by 2030, of which 160 GW will correspond entirely to renewable generation[4]. This represents a staggering 190% increase in clean energy capacities compared to the beginning of the decade[16].

Generation Technology / Vector	Target Installed Capacity (GW) in 2030	Expected Electricity Generation and Key Considerations
Solar Photovoltaic	76 GW	Represents the main growth driver. Includes a sub-target of 19 GW dedicated exclusively to distributed

		self-consumption. Along with wind, it is projected to cover two-thirds of total generation.
Wind Energy	62 GW	Includes a strategic target of 3 GW of offshore wind power, which will provide a more constant generation profile complementary to onshore wind.
Storage (Batteries and Pumped Hydro)	22.5 GW	A fundamental pillar for providing flexibility to the system. Includes lithium-ion batteries, reversible pumped hydro, and thermal storage in concentrated solar power plants.
Hydroelectric Power	14.5 GW	Maintenance and optimization of existing historical capacity, with a clear focus on converting conventional reservoirs into reversible pumped storage systems.
Renewable Hydrogen (Electrolysis)	12 GW	Installed capacity in electrolyzers, tripling the initial 4 GW target, aimed at decarbonizing heavy industry and long-distance transport.

Table 1. Installed capacity targets by technology and energy vector according to the PNIEC 2023-2030[1].

Strict compliance with these projections would guarantee, under the simulations dispatched by Red Eléctrica de España, a system capable of operating without coverage deficits or supply interruptions, sustaining a 90% share of emission-free energy when adding the contribution of

residual nuclear generation[23]. However, the ministry's own prospective models recognize the operational complexity of this mix. The simultaneous injection of wind and solar photovoltaic energy will cause recurring episodes of oversupply, forecasting energy curtailments or structural losses of up to 21.7 TWh annually, equivalent to 7.3% of the total expected renewable generation.

The Long-Term Decarbonization Strategy (ELP 2050) and Outlook to 2040

If the PNIEC represents short and medium-term tactical planning, the Long-Term Strategy for a Modern, Competitive, and Climate-Neutral Spanish Economy in 2050 (ELP 2050) operates as the State's directional compass[11]. Approved by the Council of Ministers in November 2020, the ELP 2050 establishes a binding regulatory scenario where Spain commits to reducing its Greenhouse Gas emissions by 90% compared to 1990 levels by mid-century. Residual emissions (the remaining 10%) must be entirely offset by the absorption capacity of natural carbon sinks, thus achieving the desired climate neutrality[18].

The strategy divides the transition into critical time milestones. The five-year period between 2035 and 2040 is identified as the phase of technological maturation and consolidation of flexibility infrastructures. By 2040, the electrification of the economy is expected to reach massive levels, overcoming initial resistance in complex sectors such as thermal building loads, heavy logistics, and heat-intensive manufacturing industries, through the penetration of renewable hydrogen and direct electrification systems[11].

Towards 2050, the document projects the presence of renewable energies in final energy use to reach 97%, based on a 100% renewable electricity system. To achieve this final steady state, Spain will require the sustained operation of approximately 250 GW of installed renewable capacity, which will definitively decouple economic growth from final energy consumption and reduce external energy dependence to a historic low of 13%[11].

In this prospective context to 2040, the European cross-border meshed grid will be decisive. The Ten-Year Network Development Plans (TYNDP) developed by ENTSO-E warn that achieving continental climate goals will require cross-border interconnection capacity increases of an additional 50 GW by 2030 (reaching 88 GW globally) and a supplementary 43 GW by 2040[26]. For Spain, this European imperative translates into projects of major technical complexity, such as the submarine interconnection through the Bay of Biscay. This link, which is currently 70% completed and has a budget exceeding 3.1 billion euros, will connect Gatika (Bizkaia) with Cubnezais (France) via a 400-kilometer direct current route, increasing the Iberian exchange capacity with the continent from 2,800 MW to 5,000 MW between 2027 and 2028[29].

Electricity Transmission Grid Planning (2025-2030)

No generation model is viable without the physical transmission infrastructure that forms the backbone of energy flows. Red Eléctrica de España and MITECO are processing the new Electricity Transmission Grid Planning for the 2030 horizon, designed to replace and substantially expand the current document covering the 2021-2026 period[32].

This new binding framework contemplates an unprecedented investment volume in the country's history. It is planned to raise the planned expenditure from nearly 7 billion euros in the

previous cycle to figures ranging between 13.59 billion and 17.0 billion euros by the end of the decade[34]. The justification for this capital injection lies in the flood of access and connection requests received during the study phase, where more than 5,000 proposals requested the integration of 150 GW of new photovoltaic capacity, 60 GW of wind, and over 100 GW of storage. These private sector requests far exceed the objective limits set in the PNIEC, evidencing high investor appetite that runs the risk of being frustrated by grid inadequacy[5].

To accommodate this new generation and the emergence of new industrial demand nodes, the plan projects the modernization and repowering of more than 8,160 kilometers of existing circuits, the laying of 6,706 kilometers of new power lines, and the construction of 193 new substations[34]. The underlying purpose is to maximize land use by limiting environmental impact, while averting the risk of the grid acting as an insurmountable obstacle that pushes curtailed energy levels above the profitability limits of generating plants[5].

Critical Vectors and Enablers in the Electricity System Transition

Analyzing the viability of the electricity mix at 2030, 2035, and 2040 requires going beyond the mere aggregation of megawatts of installed capacity. The massive concurrence of generation technologies with zero marginal cost and production profiles dependent on weather conditions triggers a series of market externalities and physical vulnerabilities in the transmission grid that must be addressed systemically[38].

Storage, Capacity Markets, and Adjustment Services

The PNIEC identifies energy storage as the irreplaceable pillar to sustain the 81% renewable share, setting a target of 22.5 GW of installed capacity by 2030[7]. Currently, the national storage ecosystem has approximately 8.3 GW operational, configured almost entirely by reversible pumped hydroelectric plants built in the second half of the 20th century and by thermal storage in molten salts at concentrated solar power plants[40]. The qualitative leap towards 22.5 GW in 2030 and the 30 GW projected for 2050 requires massively mobilizing private capital toward the implementation of battery energy storage systems, primarily based on lithium-ion and lithium iron phosphate chemistries[43]. The environment is favorable, given that the levelized cost of battery storage projects has plummeted globally, hitting historic lows around 78 USD/MWh in recent years, boosting the competitiveness of hybrid plants[46].

Historically, the main obstacle to storage development in Spain has been the economic problem of "missing money." This phenomenon describes the inability of capital-intensive facilities to ensure the return on their investment by operating exclusively in the day-ahead and intraday markets, due to the unpredictable volatility of arbitrage prices[40].

To correct this serious market design deficiency, MITECO approved Order TED/966/2026 in September 2026, a regulatory provision that formally creates the Capacity Market in the peninsular electricity system[48]. This mechanism regulates the financial remuneration granted to facilities in exchange for their commitment to availability (firm capacity) during periods of highest operational stress on the system. The market will operate through competitive auctions with a pay-as-bid pricing mechanism[48].

The regulations introduce a decisive incentive for new storage projects and pumped hydro plants: the possibility of signing long-term service provision contracts, lasting up to 15 years. This contractual extension provides the cash flow predictability demanded by financial institutions under *Project Finance* schemes[51]. A critical technical element that will determine each project's viability will be the "firmness coefficient" (de-rating factor) established by the system operator. This fractional parameter establishes what percentage of a battery's nominal power is recognized as remunerable firm capacity, based on its degradation and effective hours of autonomy at maximum power[53].

In parallel, storage technologies and natural gas combined cycle thermal power plants will continue to find a substantial revenue stream in the adjustment services managed by Red Eléctrica de España. The integration of growing volumes of unmanageable renewable energy increases the deviations between the generation scheduled in the forward markets and instantaneous real production. Faced with these imbalances, the system operator activates secondary and tertiary reserves to balance the grid. The importance of this mechanism is notable: in the first eight months of recent financial years, adjustment services brought in 3.375 billion euros in additional revenue for generators, with clearing prices in "upward technical constraint" services averaging 194 €/MWh, well above the arithmetic price of the day-ahead market[54].

Price Cannibalization, Curtailments, and Financial Viability (PPAs)

The dizzying speed of solar photovoltaic installation has led to the materialization of the most predictable financial risk in the Iberian energy transition: price cannibalization[57]. This economic effect arises when a vast multitude of photovoltaic parks produce and inject electricity simultaneously during the middle of the day, coinciding with maximum solar irradiation. This inelastic oversupply systematically sinks the clearing price of the wholesale market (pool), generating recurring episodes of zero or negative prices[59].

In the Iberian electricity market, the intraday price curve has adopted the morphology internationally known as the "duck curve." Depressed and flat prices are observed during daylight hours, followed by abrupt and steep spikes during sunset, the moment when photovoltaic generation disappears but residential demand reaches its zenith, forcing the dispatch of expensive gas combined cycles[59]. As a direct consequence, the "capture price" (the real average monetary value a generator obtains for each megawatt-hour fed into the grid) of photovoltaic technology has suffered severe deterioration. During 2026, the solar capture price plummeted to 31.2 €/MWh, a figure that barely represents 54% of the average wholesale market price as a whole[62].

This revenue level represents an existential threat to the long-term sustainability of the investment model. It is significantly below the standard bankability threshold (located in the range of 33 to 35 €/MWh), the lower limit required for investment vehicles to service the debt associated with building the infrastructure[63]. The situation is aggravated by the exponential increase in annual hours with zero or negative quotes (frequently exceeding 800 hours a year) and the consequent increase in technical curtailments[60].

To mitigate their exposure to extreme spot market volatility and guarantee predictable revenues, renewable developers resort to long-term Power Purchase Agreements (PPAs). However, the

deflationary pressure caused by cannibalization has also dragged down the underlying economic value of these contracts in Spain, with execution prices dangerously approaching the 30 €/MWh threshold[66]. In the current negotiations for this type of contract, the allocation of technical curtailment risk stands out as the critical variable. Determining which party bears the economic loss when the system operator orders the temporary disconnection of a park due to grid congestion defines the robustness of the generator's financial model[66].

System Stability: Synchronous Inertia, Compensators, and Grid-Forming Technology

The decarbonization of electricity generation brings with it a profound alteration of the fundamental physical properties of the grid. For decades, the frequency stability of the European interconnected system has been sustained by "synchronous inertia," a mechanical property inherently provided by the massive rotating masses of the turbines and alternators in nuclear, coal, and combined cycle plants[68]. In the event of an unforeseen generation failure or an abrupt demand disconnection, the kinetic energy accumulated in these heavy rotors instinctively dampens the rate of change of frequency, granting a vital time margin for electronic protection systems to intervene[68].

The progressive closure of coal-fired thermal plants, added to the scheduled nuclear dismantling and the extensive periods of maximum solar and wind production where combined cycles are totally or partially displaced from the pool, plunges the peninsular electricity system into a state of vulnerability due to low inertia[68]. To ward off the risk of cascading frequency drops and prevent systemic collapse episodes, the system operator has been forced to adopt unprecedented measures. In this sense, the new infrastructure planning has incorporated the installation of large synchronous compensators in key substations of the transmission grid (such as Brovales, Belinchón, Beariz, and Santa María de Grado)[71]. These massive rotating machines, which spin synchronized with the grid without providing active power, are dedicated exclusively to providing mechanical inertia, absorbing reactive power, and regulating system voltage[71].

Projecting the analysis toward the 2035-2040 decade, traditional physical rotational inertia is expected to be progressively replaced by synthetic inertia, generated through advanced power electronics algorithms. Inverters coupled to battery storage systems and renewable generation parks are undergoing a technological transition from a grid-following topology to grid-forming architectures[39]. These new converters possess the computational capacity to autonomously create and maintain a voltage and frequency reference, simulating the electromagnetic behavior of a traditional synchronous machine[39]. The recent publication by Red Eléctrica de España of technical specifications aimed at standardizing and remunerating the capabilities of these converters lays the essential regulatory groundwork for operating an electricity mix massively dominated by asynchronous power electronics[73].

New Demand Hubs: Data Centers and the Green Hydrogen Corridor (H2Med)

Any realistic projection regarding the evolution of the electricity mix in Spain for the 2030-2040 horizon must integrate the disruptive impact that new electro-intensive consumption agents will

exert, whose demand will alter the structural balances of the system. The National Artificial Intelligence Strategy initially estimated new computing power needs at around 2.5 GW, which would entail an associated electricity demand of between 3.5 and 4 GW[8].

However, the volume of access and connection rights requested and granted for this type of facility has overwhelmed any conservative estimate, exceeding 12 GW[8]. Spain, and particularly inland regions such as Aragon and the Community of Madrid, are perceived globally as optimal geostrategic locations by large tech operators, inevitably attracted by the low levelized cost of peninsular renewable energy and the high redundancy of its transmission grid[8]. In Aragon alone, project portfolios being processed contemplate aggregate investments close to 67 billion euros, involving data processing centers that will operate at very high power densities in an uninterrupted regime. This constant consumption pattern presents a profound dissonance with the intermittent generation profile of photovoltaic and wind energy, demanding colossal storage and firmness solutions[74].

In parallel, the Government reaffirms through the updated PNIEC its unwavering commitment to the development of renewable hydrogen, setting a target of 12 GW of installed capacity in electrolyzers by the end of the decade[77]. This molecular vector is the indispensable keystone for undertaking the decarbonization of industrial processes recalcitrant to direct electrification, such as the petrochemical industry, steelmaking, synthetic fertilizer manufacturing, and long-distance maritime transport[78].

The long-term commercial and industrial viability of this initiative gravitates toward the successful development of the H2Med mega-corridor. This trans-European infrastructure, consisting of the CelZa overland section and the monumental BarMar submarine interconnection, has been officially recognized as a Project of Common Interest by the European Commission. With an expected capital mobilization exceeding 2.5 billion euros and a target commercial operation date set for 2032, the system will provide the Iberian Peninsula with the logistics capacity needed to export up to 2 million tons of renewable hydrogen annually. This export volume alone represents 10% of the total estimated consumption for the European Union, effectively integrating the southern renewable production hubs with the large industrial demand centers located in France and Germany[81].

The Role of Nuclear Energy in the Transition (2027-2035)

A factor of a geopolitical, economic, and deeply technical nature that will determine the viability and shape of future scenarios is the execution of the closure schedule stipulated for the Spanish nuclear generation fleet. In 2019, the Government formalized an agreement with the sector's operating companies to carry out the definitive and phased cessation of operations of the seven commercial reactors currently in force between 2027 and 2035[85].

However, severe inflationary pressures stemming from the energy crisis unleashed following the outbreak of the conflict in Ukraine, the enormous volatility in liquefied natural gas markets, and the pressing technical need to maintain an emission-free synchronous generation base motivated a partial reconsideration of this policy. In 2026, the Executive, backed by a favorable technical opinion issued by the Nuclear Safety Council (CSN), exceptionally authorized an

extension of the useful life of the Almaraz nuclear power plant, postponing the initial shutdown scheduled for 2027 and 2028 until a date no earlier than June 2030[87].

The heated debate over the potential extension of the operational life of the rest of the national nuclear fleet (which includes key facilities such as Cofrentes, Ascó, Vandellós, and Trillo) beyond the 2035 time boundary radically divides the actors in the energy sector, shaping two antagonistic positions[86]:

Arguments for Extension (Resilience and Competitiveness): Industry representative entities, supported by prospective models developed by strategic consulting firms, argue that preserving installed nuclear capacity prevents an undesirable structural dependence on natural gas during valleys of renewable production. Economic studies estimate that executing a premature closure during crises in fossil fuel markets would have inflated the aggregate electricity bill by nearly 8.8 billion euros annually[92]. In the long term, it is calculated that postponing the closures could yield structural savings of between 14 and 15 €/MWh in the energy costs borne by the electro-intensive industry[93]. Additionally, from a strictly operational level, they argue that the synchronous inertia and continuous base load provided by these reactors drastically mitigate the technical need to resort to the costly adjustment services operated by Red Eléctrica de España[93].

Arguments Against Extension (Inflexibility and Hidden Costs): In open opposition, platforms and organizations defending the ecological transition, such as the Fundación Renovables, warn that the artificial prolongation of the useful life of an obsolete nuclear fleet would act as a structural bottleneck in the transmission grid, severely limiting evacuation capacity and hindering the deployment of new renewable plants and electrochemical storage systems[96]. They correctly point out technically that the extreme rigidity inherent in fission reactors (whose power modulation against wind and solar generation peaks is slow and limited) substantially aggravates the phenomenon of price cannibalization and multiplies the volumes of curtailed energy[95].

In the strictly economic sphere, detractors calculate that a generalized moratorium on the closure would impose an extra cost exceeding 3.831 billion euros on final consumers over the decade, in addition to profoundly altering the financial provisions and long-term technical planning managed by Enresa under its 7th General Plan for Radioactive Waste and the future dismantling work on the facilities[96]. Added to this, the levelized cost of energy associated with carrying out the massive investments required to keep reactors designed in the last century operational is increasingly asymmetric compared to the overwhelming competitiveness of the synergistic combination between state-of-the-art photovoltaic parks and advanced battery storage systems[101].

Prospective Scenarios for the Future Energy Mix (2030, 2035, 2040)

After aggregating the analysis of the regulatory environment, the real dynamics of short-term markets, the development of the levelized cost of technologies, and the capacity boundaries of the physical infrastructure, it is imperative to project future models. Below, three divergent scenarios are proposed regarding the evolution of the Spanish electricity mix over the next two

decades, assuming feasible variations in the degree of technological adoption and the international macroeconomic environment.

Scenario 1: Accelerated Transition, Storage Consolidation, and Regulatory Compliance (The PNIEC / ELP Scenario)

This prospective model assumes a highly successful execution of the current master plans, backed by a geopolitical and institutional climate that mitigates bureaucratic obstacles and fosters the uninterrupted cost reduction of the transition's enabling technologies.

The trigger for this scenario lies in the resounding success of the Capacity Market established by Order TED/966/2026. Periodic auctions manage to assign fair and financially attractive firmness coefficients to battery storage systems. This long-term price signal (contracts of up to 15 years) allows developers to structure large-volume corporate financing operations, attracting institutional capital. Simultaneously, the aggressive 2025-2030 Transmission Grid Planning overcomes environmental bottlenecks, guaranteeing the timely construction of new substations and the reinforcement of critical lines, thereby avoiding dreaded system congestion.

Reaching the 2030 milestone, the Spanish State consolidates the installation of the 160 GW of renewable power required by the PNIEC. The massive and capillary deployment of storage systems (which reach the 22.5 GW mark by combining the momentum of batteries and the execution of complex pumped hydro projects co-financed by European funds) successfully flattens the duck curve. The vast amount of photovoltaic energy generated at midday is efficiently absorbed by storage systems and reinjected during nocturnal demand peaks. This dynamic restores balance to the wholesale market, progressively raising solar capture prices back to profitable thresholds, thus restoring confidence and bankability in the long-term PPA markets.

In this context of demonstrated renewable solidity and flexibility, the Almaraz nuclear power plant proceeds to close its reactors in 2030 in an orderly manner, with its base generation void being seamlessly filled by dispatchable storage systems and a rejuvenated wind fleet, initiating the closure program for the rest of the national nuclear fleet.

At the midpoint of the decade, around 2035, the deep electrification of economic demand is an undeniable fact. Light mobility is electrified to a significant percentage, and the decarbonization of the heating sector via heat pumps dominates urban environments. The nuclear phase-out schedule progresses without alterations at the Ascó, Cofrentes, Vandellós, and Trillo plants. Natural gas and large combined cycle plants are strictly relegated to a strategic reserve function or a capacity mechanism of last resort, dispatching energy in less than 5% of the hours of the year. The dynamic stability of the grid is routinely ensured through a massive integration of electronic converters equipped with grid-forming functionalities and the continuous operation of new synchronous compensators, operating in perfect coordination.

By the 2040 horizon, Spain emerges as an interconnected and resilient system, operating in a quasi-steady state with an electricity mix that provides 90-95% completely decarbonized and autonomous energy. The H2Med submarine corridor constantly transports the projected volumes of green hydrogen towards France and Central Europe. In turn, floating foundation offshore wind technology reaches commercial maturity, populating the Atlantic, Cantabrian, and Canary Island marine demarcations, injecting a stable generation profile that minimizes

dependence on solar irradiation cycles. The country capitalizes on a competitive energy model, attracting high value-added industrial investments in fields like artificial intelligence and intensive data processing under a green guarantee seal.

Scenario 2: Technological Bottleneck, Grid Delays, and Nuclear Extension (Constant Thermal Backup Scenario)

This scenario models a suboptimal but highly plausible situation, derived from the failure to scale flexibility solutions at the grid level and the administrative logjam in physically expanding electricity transmission infrastructures.

The degenerative dynamic of this scenario originates from critical implementation failures in the regulatory and macroeconomic framework. The incipient capacity market proves insufficient; the remuneration guarantees or the imposed firmness coefficients are conservative, disincentivizing large investment funds, which reclassify utility-scale battery projects as high-risk assets. In parallel, geopolitical tensions over the control of critical mineral supply chains (lithium, cobalt, copper) cause an unexpected increase in the levelized cost of storage. As a fatal consequence, solar price cannibalization worsens year after year, chronically sinking capture prices below 25 €/MWh. Developers' finances collapse, triggering a wave of painful refinancing, sector consolidations, and the absolute paralysis of new private developments under the merchant modality.

By 2030, the balance is discouraging. The country misses the ambitious metrics of the updated PNIEC by a wide margin. Photovoltaic capacity stagnates, and the volume of operational storage barely increases due to the lack of incentives. The technical inability to evacuate renewable generation surpluses during daylight hours through a transmission grid suffering a structural delay in its expansion drives curtailment rates above 15%.

Under the imminent pressure of suffering critical synchronous generation deficits in winter periods and faced with the prospect of an unsustainable rise in backup natural gas prices, a new political, economic, and regulatory consensus determines the need to undertake a massive and generalized extension of the Spanish nuclear fleet's useful life. The definitive dismantling operations at Almaraz are postponed indefinitely, and the other sites like Cofrentes, Ascó, and Vandellós are renewed and upgraded through public-private capital injections to operate for another fifteen or twenty years, absorbing the increases in Enresa's waste management fees.

Advancing towards 2035, the peninsular electricity system operates under a chronically stressed dual paradigm. The nuclear fleet continues to provide 7 GW of uninterrupted baseload power. While this guarantees emission-free supply, its total inflexibility to fluctuations dramatically exacerbates the curtailment of wind and solar energy during peak production months, further penalizing independent renewable producers. To cover the variability during sunsets and demand peaks, dependence on combined cycles remains high. With CO_2 emission allowance prices systematically exceeding 150 €/ton under the toughened European ETS II scheme, the marginal price of the day-ahead market experiences episodes of enormous volatility and price spikes when gas sets the final price, penalizing the competitiveness of small businesses.

In 2040, the state of the grid demands a rethink. The transition, weighed down for a decade, must be reactivated through binding impositions and regulations from European community

directives. Spain has partially squandered its initial competitive advantage over Northern Europe. The definitive dismantling of the conventional nuclear fission era is relegated without a concrete timeframe to the second half of the century, and the climate neutrality objectives outlined in the ELP 2050 face a very high risk of structural non-compliance.

Scenario 3: Asymmetrical Demand Explosion and Market Polarization (Dissociated Electro-Intensive Scenario)

This third prospective model projects a future where electricity demand growth, catalyzed by global digitization and accelerated reindustrialization based on hydrogen, outpaces the organic growth capacity of the country's generation and transmission infrastructures.

The engine of this scenario is the massive influx of capital from hyperscalers and international tech consortiums, which land on the peninsula attracted by the combination of talent, geographic location, and a mirage of infinite availability of cheap energy. Data processing centers and artificial intelligence infrastructure materialize the more than 12 GW of requested capacity. In parallel, supported by the deployment of vast European funds associated with the Recovery, Transformation, and Resilience Plan (PERTE for Industrial Decarbonization), heavy national manufacturing industry and energy consortiums build giant electrolyzer clusters to dominate the nascent renewable hydrogen market.

By the 2030 horizon, the outlook is one of systemic overload. Total electricity demand undergoes an upward shock that shatters all models foreseen in the PNIEC, easily crossing the 400 TWh per year threshold. However, the renewable installation effort, despite approaching theoretical capacity targets, proves inadequate in terms of the constant production profiles or baseload required by new customers. Furthermore, heavy environmental permitting bureaucracy slows down the materialization of the reinforcements planned in the 2025-2030 Electricity Planning, permanently saturating key nodes. Even though fundamental international interconnections like the Bay of Biscay link manage to come online, the flow dynamic counter-intuitively reverses: instead of acting as an export route for diurnal Iberian photovoltaic surpluses, the country is forced to import massive flows of nocturnal baseload energy (frequently of French nuclear origin) to keep the monstrous data centers and electrolyzers operating, altering an international exchange balance that had been structurally export-oriented.

Around 2035, this structural mismatch between supply and demand morphology solidifies a deeply polarized electricity market. Large tech consortiums and green hydrogen producers opt for financial shielding strategies, formalizing complex physical Power Purchase Agreements that monopolize and sequester the extraction rights of all new onshore wind installations, concentrated solar power parks, and photovoltaic generation complexes hybridized with large battery parks of 4 to 8 hours duration. In contrast, traditional residential, commercial, and industrial consumers, lacking the financial leverage to secure these types of private long-term contracts, are exposed to the behavior of a day-ahead market diminished in its liquidity of cheap energy and highly dependent on the marginal cost dictated by emergency imports and the punctual operation of combined cycles.

Towards 2040, the immense pressure and profitability associated with providing a highly uninterrupted capacity factor energy supply triggers the spectacular commercial takeoff of offshore wind on the deep platforms of the Atlantic, Cantabrian, and Mediterranean, establishing

Spain as a technological benchmark in energy-oriented naval engineering. The international BarMar hydrogen pipeline operates at its maximum planned capacity. However, faced with the insufficiency of national renewable resources to keep pace with this hyper-electrified economy, the Iberian Peninsula assumes a mixed logistical role, acting as a great processor and transient importer of hydrogen and synthetic derivatives coming from North Africa, which are refined and pumped through European pipelines towards the central Franco-German corridor.

Conclusions

The ongoing transformation and planned restructuring of Spain's energy system over the 2030-2040 horizon, quantified in the National Integrated Energy and Climate Plan and conceptualized in the Long-Term Decarbonization Strategy, undoubtedly constitute the vastest and most complex critical infrastructure reengineering process in contemporary national economic history.

The technical consolidation and effective materialization of an electricity mix that meets international commitments (reaching an 81% renewable share in 2030 and orbiting towards 100% emission-free coverage by mid-century) do not solely rely on the arithmetic ability to annually install volumes of raw photovoltaic or wind power. Deep analysis of current macroeconomic and engineering variables demonstrates unequivocally that the entire model's success will pivot on the early and effective resolution of the systemic flexibility bottleneck afflicting the market and the grid.

In the absence of a rapid mobilization of capital that allows the deployment of the required 22.5 GW of interday and seasonal storage, the transmission grid will show signs of irreversible physical congestion. The financial corollary of this inability will be the unbridled increase in curtailed energy and the perpetuation of solar cannibalization, a phenomenon that relentlessly destroys the private investor's operating margin and disincentivizes the provision of new funds in a vital sector. Therefore, the expeditious development and successful financial design of the new Capacity Market (Order TED/966/2026), complemented by the technical obligation to incorporate advanced electronic converters with grid-forming capability and robust synchronous compensators, act as the true keystones that will sustain the system's technical and commercial viability in the coming decades.

Simultaneously, policymakers will have to orchestrate an extremely complex balancing act in the coming five-year periods: harmonizing a contractually stipulated nuclear capacity phase-out process with the pressures of a new industrial era. The explosion of electricity demand derived from Artificial Intelligence and the pan-European consolidation of strategic hydrogen corridors, such as H2Med, will strain any static demand forecasting scenario. Ultimately, depending on the degree of execution, bureaucratic agility, and technological maturity that storage achieves during this critical five-year period, Spain will indisputably position itself as a highly competitive global exporter of renewable energy (Scenario 1), succumb to delay and congestion dynamics that will force a conservative return to the old thermo-nuclear backup model (Scenario 2), or mutate into an electro-intensive hyper-consumption economy that is polarized and technologically dissociated from the traditional market (Scenario 3).

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